

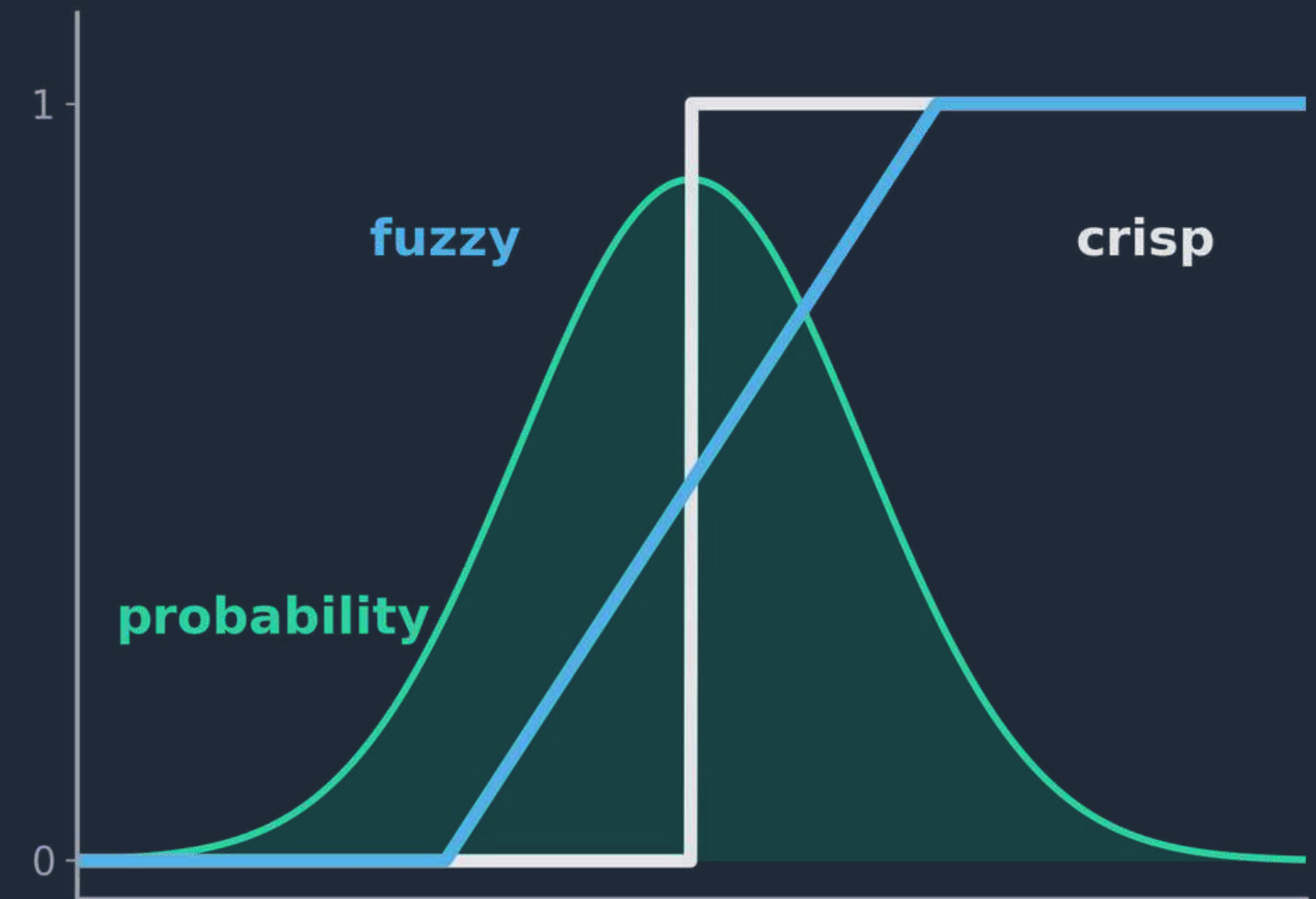
Advanced AI and Uncertainty

Four kinds of imperfect information, two ways to model vagueness, and probability for chance and missing data

Ahmed Ayman Tayel

Registration ID 232611000001

College of Computers and Artificial Intelligence, Matruh University



Crisp logic, fuzzy sets and probability: three tools, three kinds of uncertainty



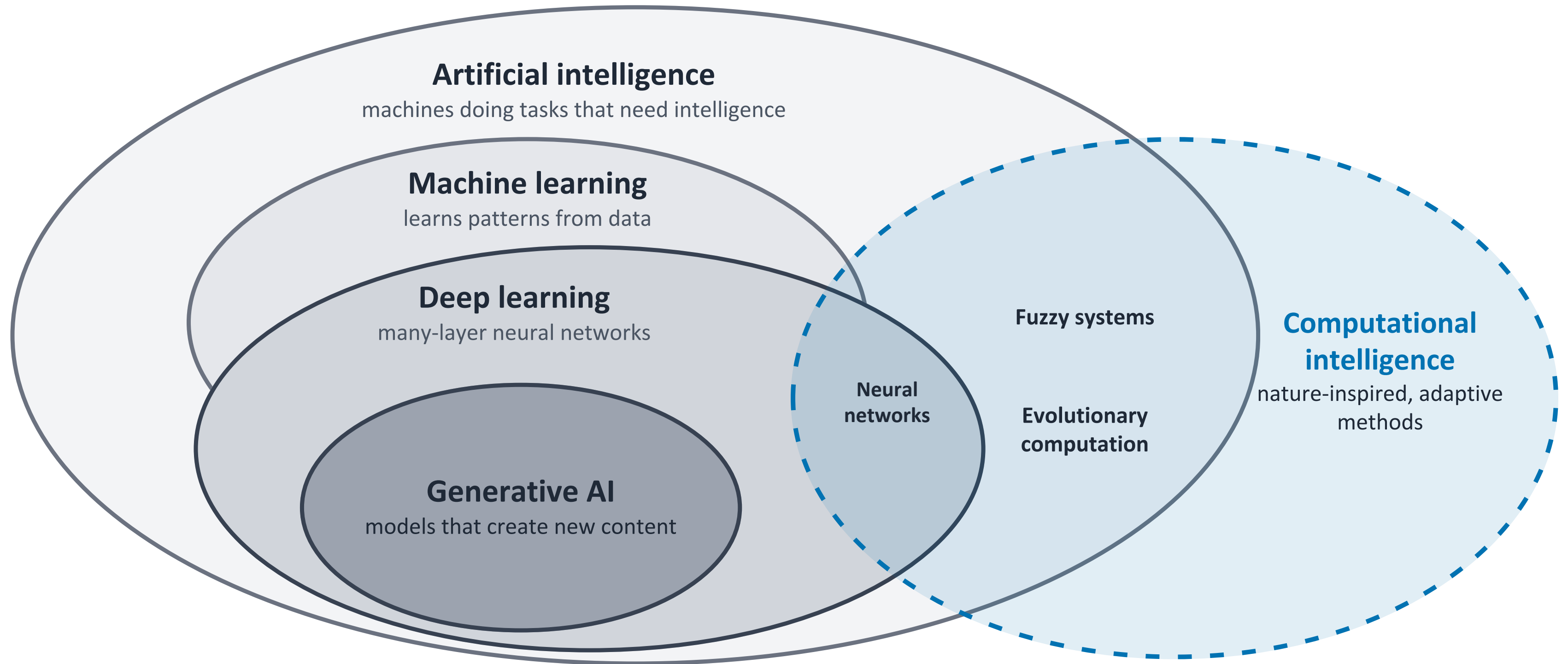
Agenda and roles

Act	Part	Topic	Slides
Act 1	1	The AI landscape today	3–5
	2	Why uncertainty matters in AI	6–8
Act 2	3	Four kinds of imperfect information	9–10
	4	Aleatoric vs epistemic uncertainty	11–13
Act 3	5	Crisp logic and its limits	14–16
	6	Fuzzy logic: the basic idea	17–19
Act 4	7	Probability and missing data	20–23
	8	Live demo and link to the course project	24–27

Presenter : Ahmed Ayman Tayel | Registration ID: 232611000001 | Team 01

The live Google Colab demo (about 4 minutes) runs between slides 24 and 25.

AI is the umbrella; each ring is a narrower idea



Computational intelligence overlaps AI and machine learning; sources draw its boundary differently.

Two ways to build AI: write the rules, or learn from data

	Symbolic (knowledge-based)	Data-driven
Knowledge comes from	Experts write rules and facts	A model is trained on examples
It decides by	Logical inference over symbols	Numeric computation with learned parameters
Strength	Explainable; works with little data	Copes with noisy input such as images
Weakness	Brittle and costly to write	Needs much data; opaque; can be confidently wrong



Real example: MYCIN (1970s)

If-then rules with certainty factors advised on antibiotics for blood infections.



Real example: retinopathy AI

Deep learning trained on about 128,000 retinal photographs flags the disease.

Four classic AI problems, each harder under uncertainty



Search

Find a sequence of actions that reaches a goal.

Where uncertainty enters: The map, or the opponent's next move, may differ from the model.



Knowledge representation

Encode facts so a program can use them.

Where uncertainty enters: Facts are incomplete or only mostly true.



Planning

Choose actions to reach goals over time.

Where uncertainty enters: Actions have unpredictable effects and the world changes.



Reasoning under uncertainty

Draw sound conclusions from partial, noisy information.

Where uncertainty enters: Needs probability or fuzzy tools: the rest of this course.

Uncertainty has four common sources



Noisy sensors

Two thermometers give two readings for the same patient.



Human language

“High risk” means different things to different doctors.



Incomplete knowledge

A test was never ordered; the cause is only partly known.



A changing world

Patients and hospitals change, so last year’s model drifts.

These four are different problems; Part 3 separates them because each needs a different tool.

Ignoring uncertainty makes a system confident and wrong

POINT ANSWER

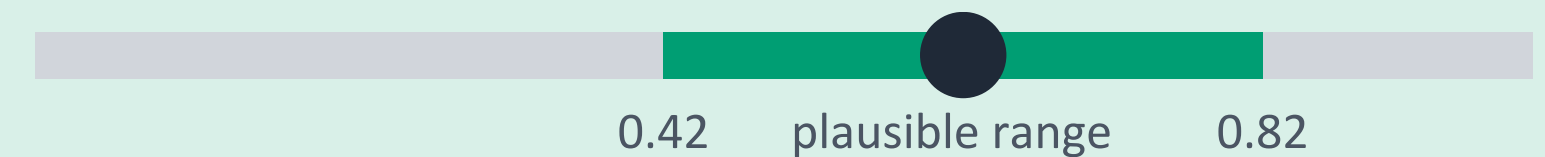
Diabetes: YES

Confidence: 99 %

No range, no warning, no missing-data flag.

ANSWER WITH UNCERTAINTY

Risk 0.62

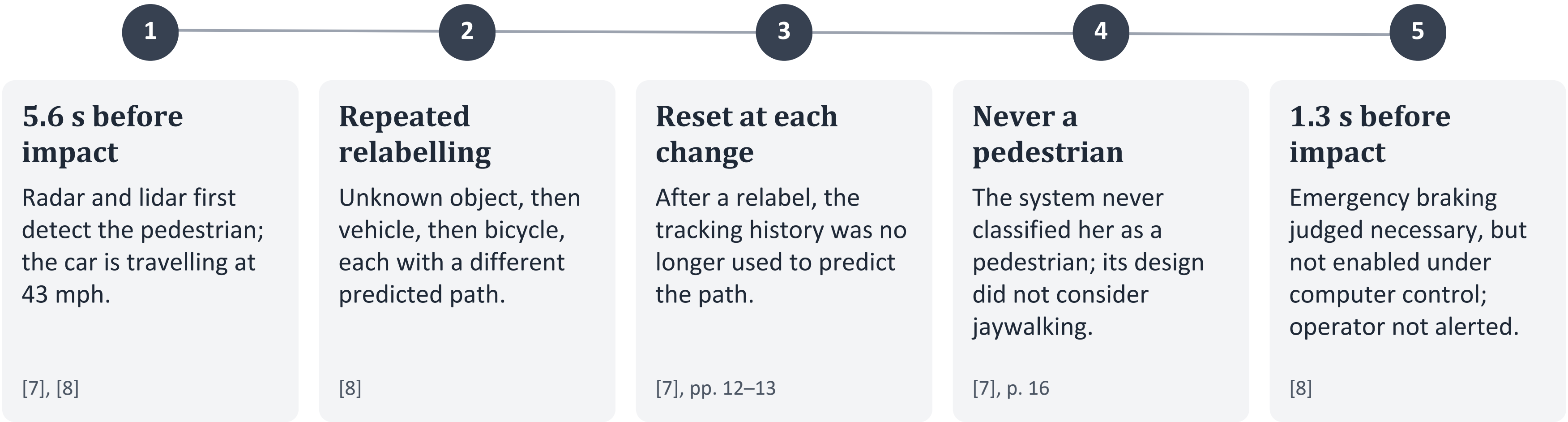


Glucose not recorded
Suggest: repeat the test

Illustrative example, not real data

- 1 A classifier must answer, even when the input is unlike its training data.
- 2 A model's score is not automatically a calibrated probability; modern deep networks are often over-confident [6].
- 3 The user then trusts a number that hides what the system does not know.

Case study: a self-driving car and an object it could not classify



NTSB probable cause: the operator’s failure to monitor the road (visual distraction). The report also documents how the system handled the uncertain detection.

Our reading: uncertainty was reset, not carried forward, and there was no safe fallback.

Related: a retinopathy AI in Thai clinics struggled with imperfect image quality [9].

Four kinds of imperfect information



Randomness

The outcome varies by chance.

Example: A dice roll; tomorrow's rain



Vagueness

The concept has no sharp boundary.

Example: "Tall", "hot", "high risk"



Imprecision

The value is known only within a range.

Example: A sensor reading of 30 ± 2 °C



Missing data

The value is not recorded at all.

Example: An empty field in a patient record

One patient record contains all four kinds

Patient A, diabetes screening visit (illustrative record)

Kind	Example in Patient A’s record	What is uncertain	Fixed by measuring more or better?
Randomness	Will A develop diabetes within five years? Risk estimate: 20 %	The outcome itself	No. Even with perfect data the outcome stays a matter of chance.
Vagueness	Is A “high risk”?	Where “high” begins	No. A better instrument does not define “high”; a definition or a degree of membership does.
Imprecision	A’s temperature is 37.4 ± 0.3 °C	The exact value	Partly. A better instrument narrows the range, but never to zero.
Missing data	A’s insulin field is empty	Whether a usable value exists	Yes, if the test can still be done; otherwise it must be handled (Part 7).

Two types of uncertainty: chance in the world, and gaps in our knowledge

Aleatoric



MEANING

Variability built into the data

MORE DATA?

Does not remove it

MEDICAL EXAMPLE

Two patients with identical glucose, age and BMI respond differently to the same treatment.

WORD ORIGIN

Latin alea, a die

Epistemic



MEANING

Lack of knowledge or data

MORE DATA?

Can reduce it

MEDICAL EXAMPLE

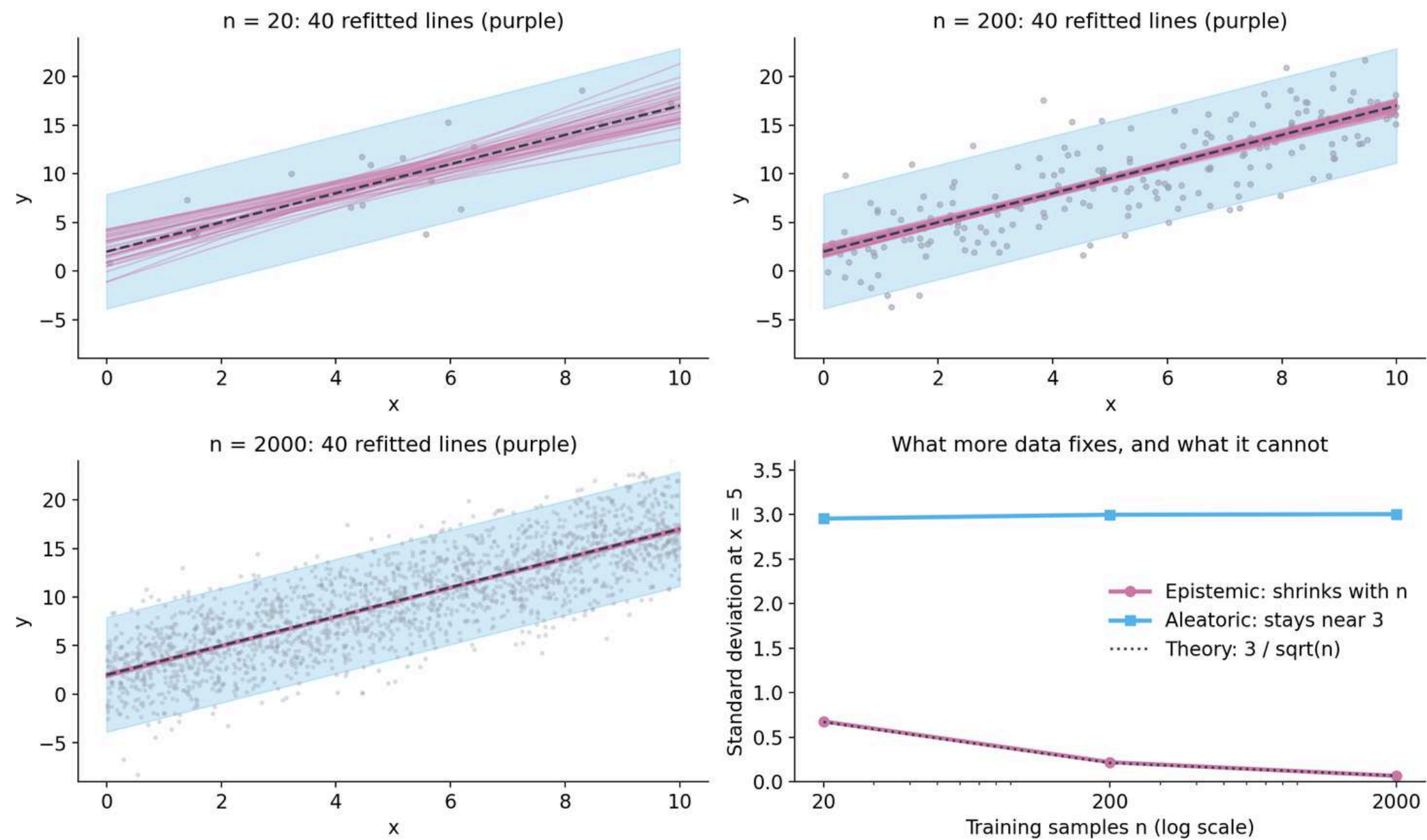
The model has seen only a handful of patients over 70, so its estimate for them is shaky.

WORD ORIGIN

Greek episteme, knowledge

More data shrinks model doubt, but not noise

Simulated data: x = today’s fasting glucose, y = next month’s value; the same straight-line model is fitted on 20, 200 and 2,000 patients.



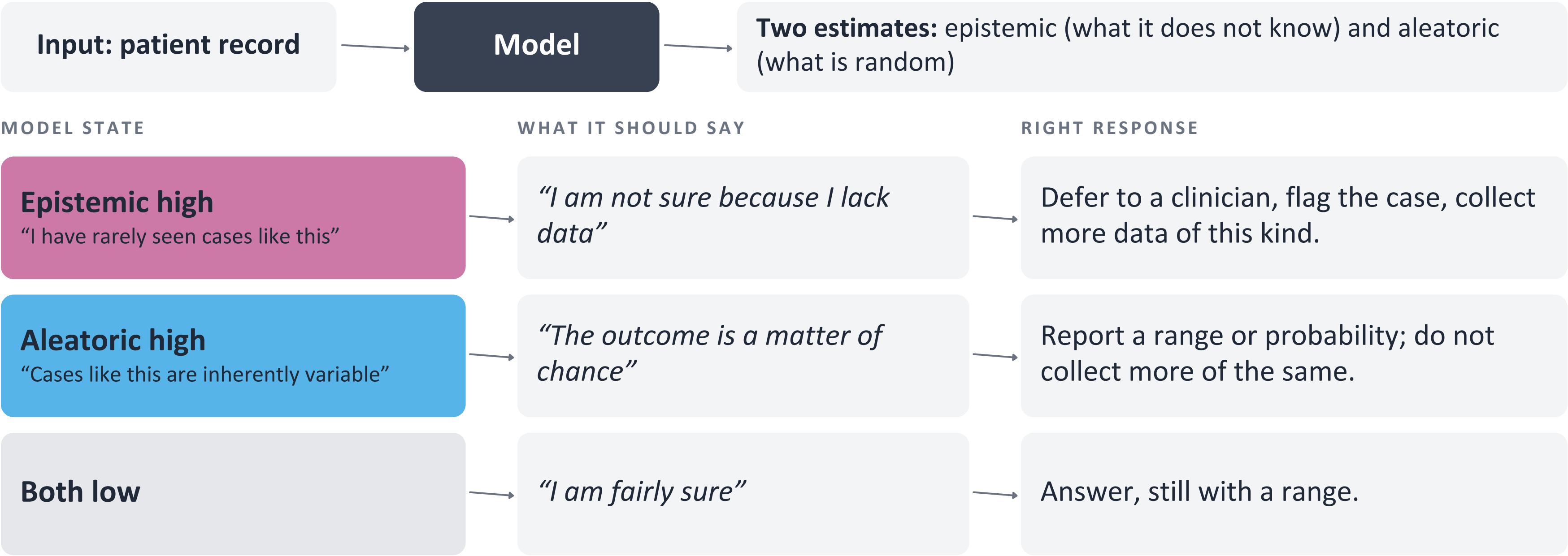
Blue band = irreducible noise (about 95% of data); dark dashed line = true relationship. Simulated data.

Blue band (aleatoric): patients with the same reading still differ by chance. Noise sd stays about 3 at n = 20, 200 and 2,000.

Purple lines (epistemic): the model’s own doubt. Spread of its prediction at x = 5: 0.68, then 0.22, then 0.07 as n grows.

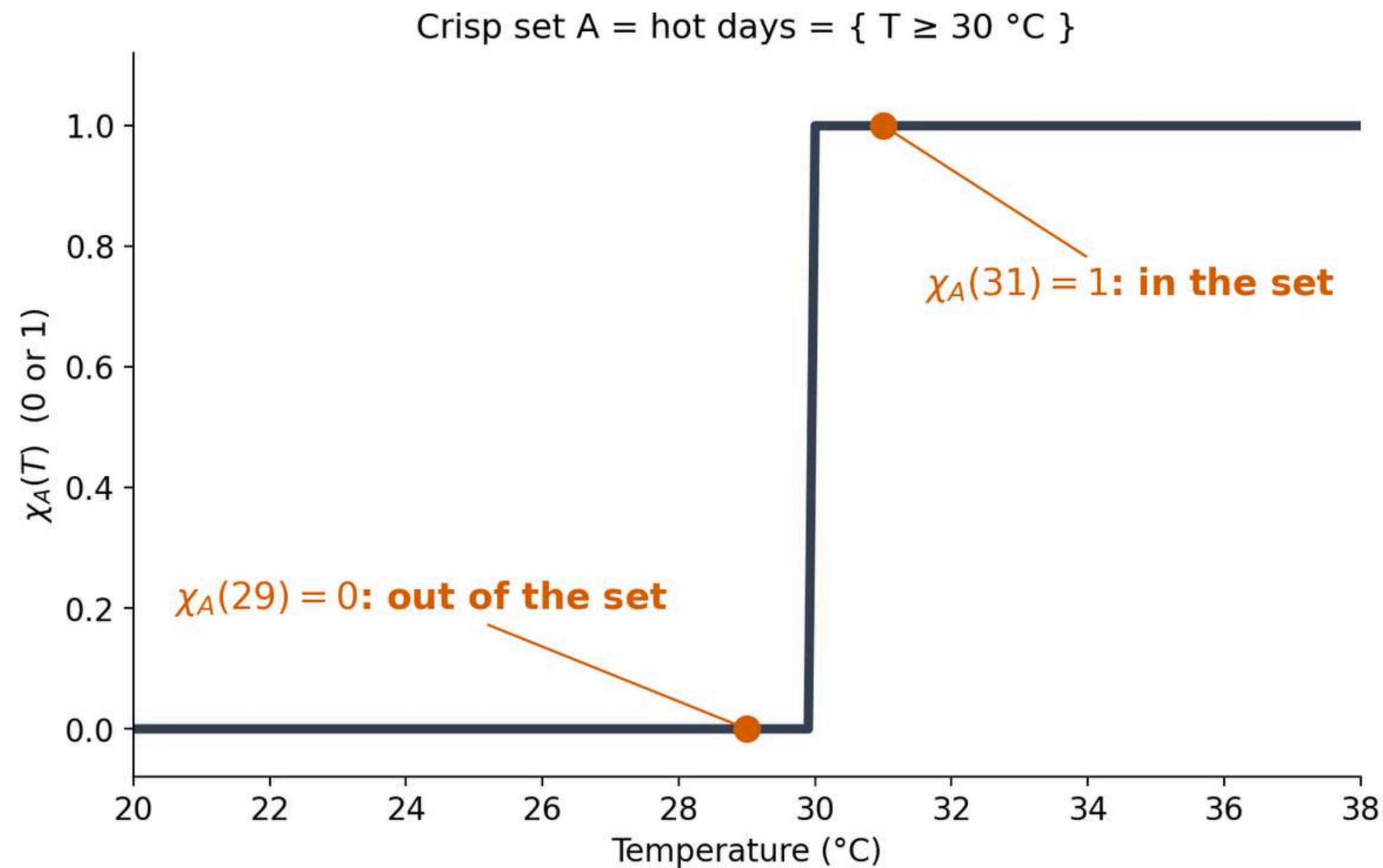
More data fixes the purple part, not the blue part.

Which kind of doubt it is decides what to do next



Wrong diagnosis of the doubt means wrong action: collecting data for pure noise is wasted effort; averaging away ignorance hides risk.

Crisp logic: every element is in (1) or out (0)



A crisp set A has a characteristic function:

$$\chi_A(x) = 1 \text{ if } x \in A$$

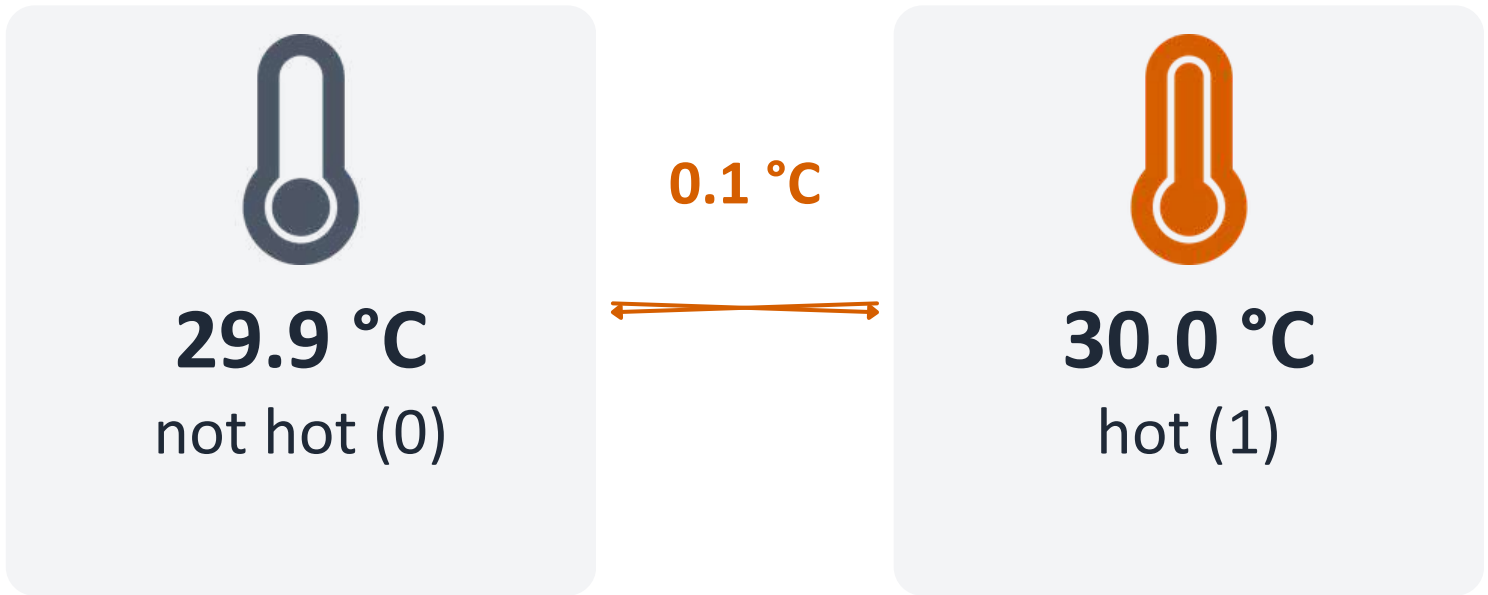
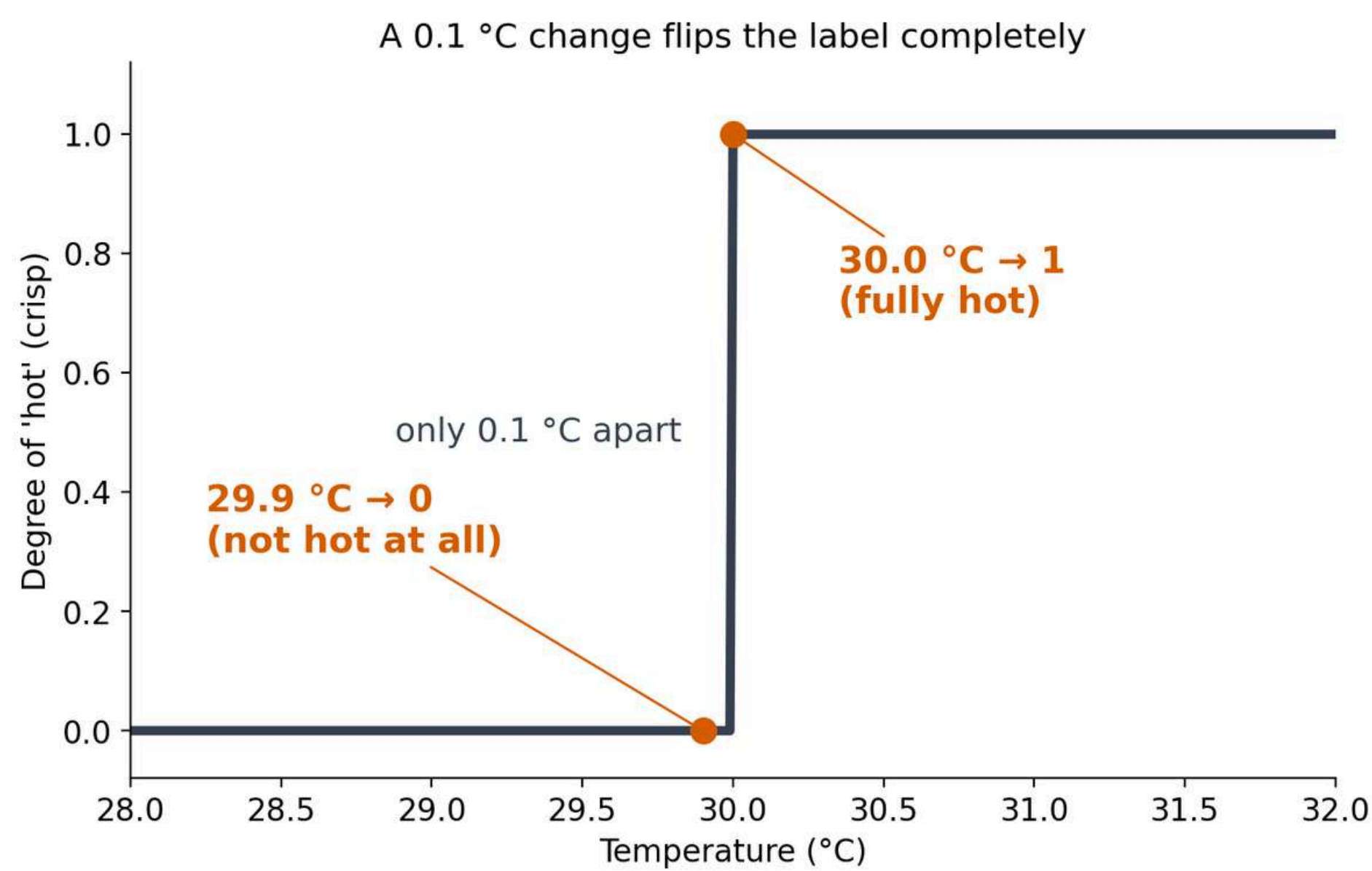
$$\chi_A(x) = 0 \text{ if } x \notin A$$

$$A = \text{hot days} = \{ T : T \geq 30 \text{ °C} \}$$

$$\chi_A(31) = 1, \quad \chi_A(29) = 0$$

**Only two truth values: true or false.
Membership is all or nothing.**

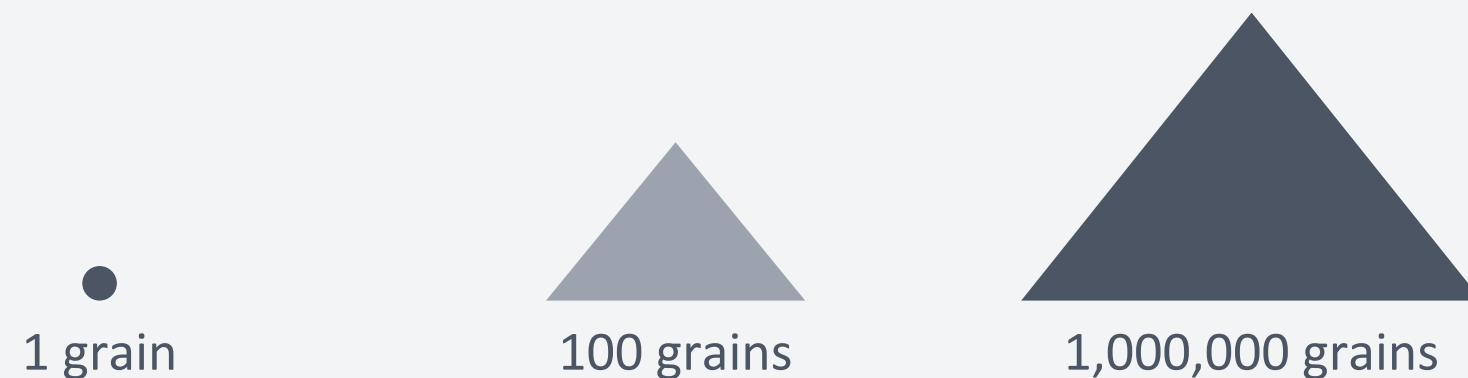
If “hot” starts at 30 °C, then 29.9 °C is not hot at all



- 1 A change of 0.1 °C flips the label completely.
- 2 Thermometers are imprecise (± 0.3 °C), so noise alone can flip the label.
- 3 Real concepts (“tall”, “high risk”) do not change abruptly at one point.

Vague concepts break crisp logic; sharp rules still have a place

The sorites (heap) paradox



1. One grain of wheat does not make a heap.
2. If n grains do not make a heap, adding one grain does not either.
3. Therefore one million grains do not make a heap.

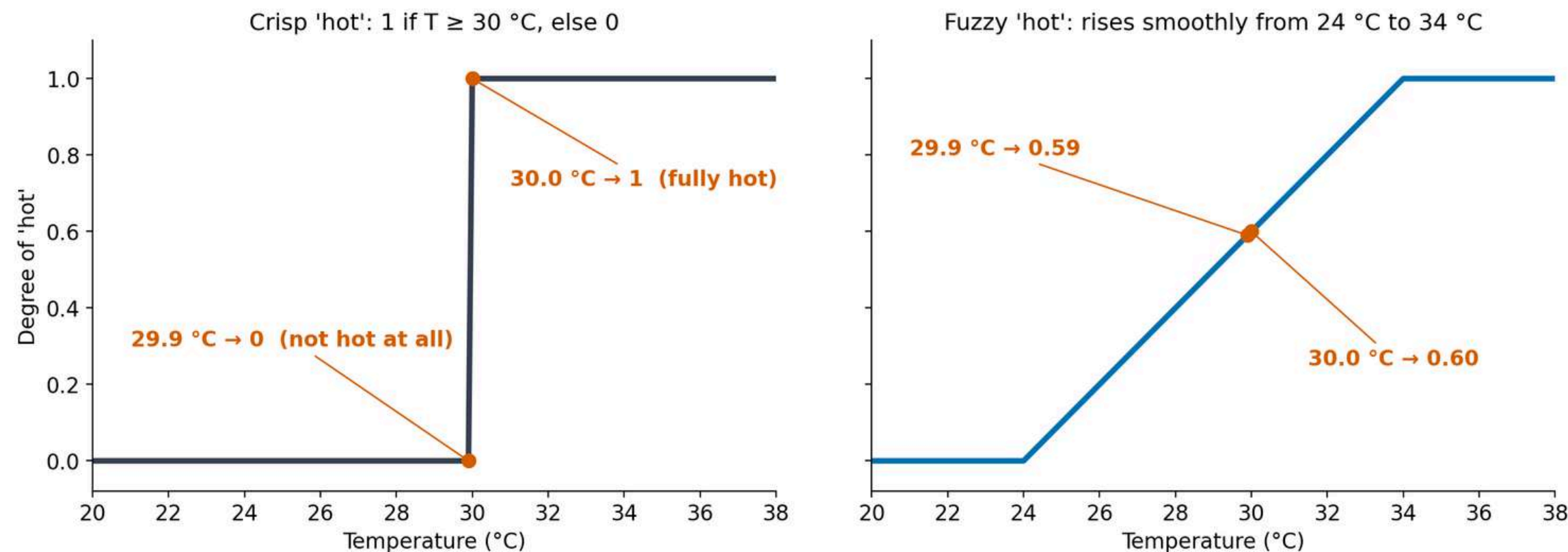
Same puzzle for tall, hot, high risk.

When crisp is still the right choice

- ✓ Legal age: 18 is the line, by law
- ✓ Pass/fail defined by regulation
- ✓ On/off safety interlocks
- ✓ Exact inputs, e.g. a date of birth

Use crisp when a rule defines the boundary and a clear line is worth more than a smooth one.

Fuzzy sets soften the edge: membership is a degree from 0 to 1



A fuzzy set A has a membership function $\mu_A(x)$ between 0 and 1: the degree to which x belongs to A . A crisp set is the special case where only 0 and 1 occur.

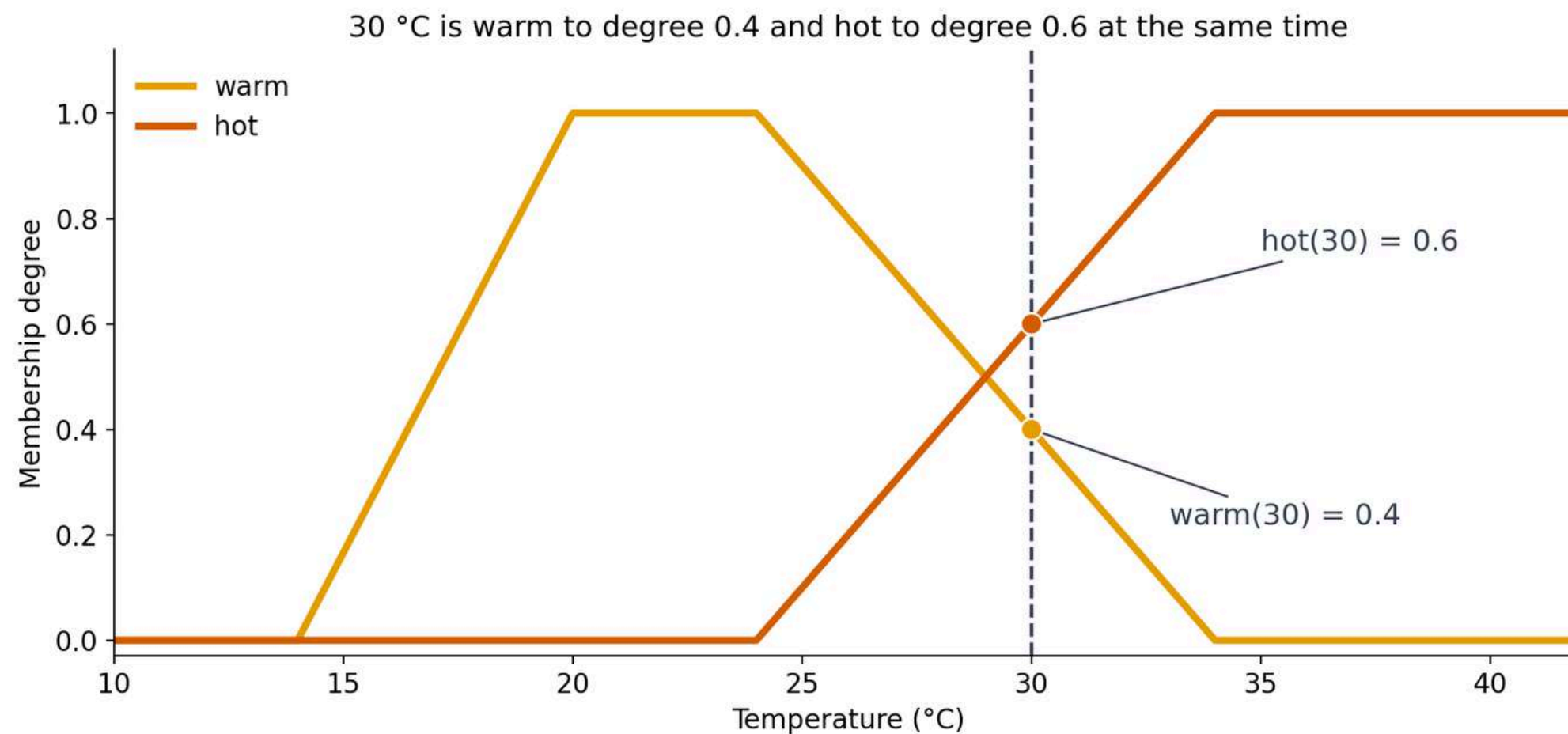
Reading the plot

Left: crisp “hot” jumps from 0 to 1 at 30 °C.

Right: fuzzy “hot” rises smoothly from 24 to 34 °C; 29.9 gives 0.59 and 30.0 gives 0.60.

L. A. Zadeh, “Fuzzy sets,” *Information and Control*, 1965: a class of objects with “a continuum of grades of membership”.

30 °C is warm to degree 0.4 and hot to degree 0.6



```
hot(T)  = (T - 24) / 10  
         clipped to 0 .. 1  
warm(T) = trapezoid  
         (34 - T) / 10
```

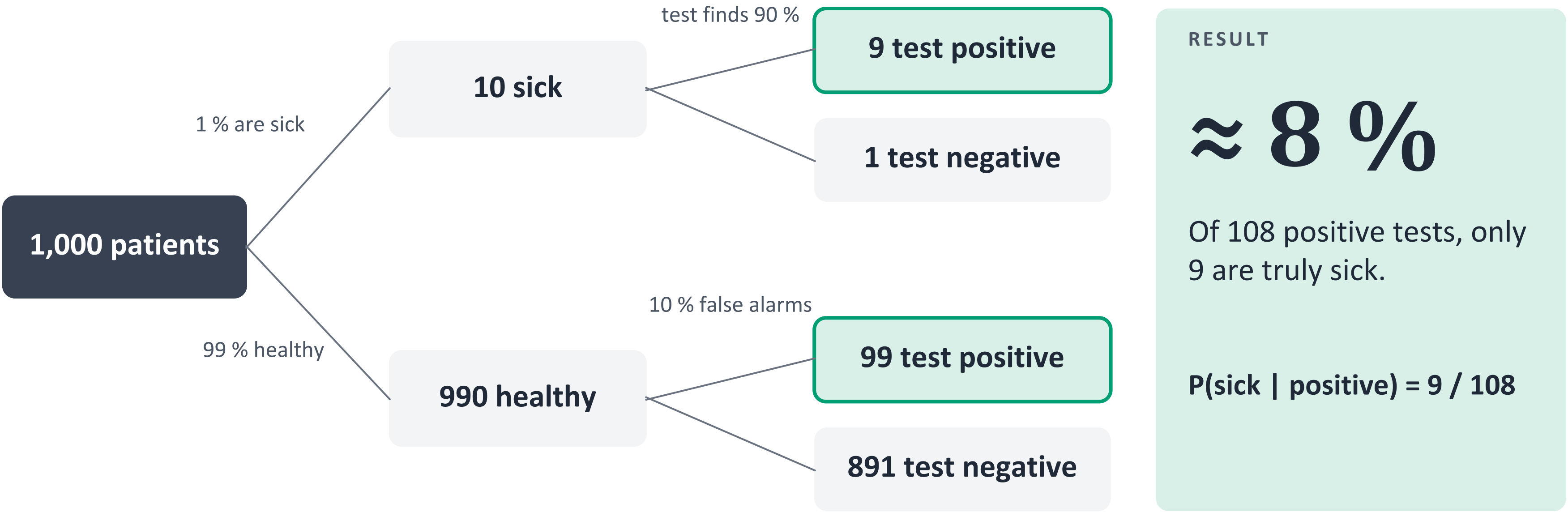
```
hot(30)  = (30-24)/10 = 0.6  
warm(30) = (34-30)/10 = 0.4
```

One temperature, two sets, two degrees. A crisp rule would force a single label.

The shapes are design choices, usually taken from expert judgement or data.

A positive test does not mean 90 % sure

Illustrative numbers, not real data



CONDITIONAL PROBABILITY
 $P(A \mid B) = P(A \text{ and } B) / P(B)$

BAYES' THEOREM, AS INTUITION
 $P(\text{sick} \mid +) = P(+ \mid \text{sick}) \times P(\text{sick}) / P(+)$

Read “|” as “given”. Full treatment of Bayes’ theorem and Bayesian networks: Week 4.

Why a value is missing matters as much as how much is missing

MCAR

missing completely at random

The chance of being missing is unrelated to any value, observed or not.

Medical example

A lab tube breaks at random, so that patient's glucose is lost.

MAR

missing at random

The chance of being missing depends only on values that were observed.

Medical example

Older patients (age is recorded) are less often sent for a fasting glucose test.

MNAR

missing not at random

The chance of being missing depends on the missing value itself.

Medical example

Patients with very high glucose feel too ill to attend the follow-up, so exactly the high values are absent.

Observed data can suggest MCAR vs MAR, but MAR vs MNAR cannot be verified from the observed data alone.

Every fix for missing data has a price

Strategy	What it does	Risk it introduces	Evidence from the demo
Deletion	Drop every row with a missing value	Loses information; biased unless data are MCAR	Keeps only 392 of 768 patients (51.0 %)
Mean or median imputation	Fill gaps with the column mean or median	Shrinks variance; treats a guess as a measurement	Insulin spread falls from sd 118.8 to 85.0
Missing-value indicator	Add a 0/1 column “was missing”	Keeps the fact of missingness, but the filled value is still invented; adds columns	374 patients flagged for insulin

Each fix silently assumes something about why the data are missing.

Three tools, three different questions

	Crisp logic	Fuzzy logic	Probability
Question it answers	Is x in the set: yes or no?	To what degree does x fit the concept?	How likely is the event, given what I know?
Type of uncertainty handled	None: assumes sharp concepts and exact facts	Vagueness	Randomness and lack of knowledge
Suitable application	Legal-age check; regulatory pass/fail	Comfort or risk labels such as “hot” or “high risk”	Diagnosis from test results; weather forecast
Limitation	Jumps at the threshold; no degrees	Degrees depend on chosen membership functions; not a measure of chance	Needs data or priors; cannot define a vague word by itself

Live demo: real patient data, three questions



Pima Indians Diabetes Database

768 women aged 21 or older

8 measurements + diabetes outcome

NIDDK; Smith et al. (1988)

A

Missing values

Count them, and find the hidden zeros.

B

Crisp vs fuzzy

Same variable, two definitions of “obese”.

C

More data

What does a bigger sample fix, and what not?

Notebook: W01_UncertaintyDemo_Team01.ipynb

Runs top to bottom in Google Colab in about 30 seconds.

Colab share link here :

https://colab.research.google.com/github/ahmedtayel714/Advanced-AI-and-Uncertainty--W1/blob/main/W01_UncertaintyDemo_Team01.ipynb



Three lines from the demo

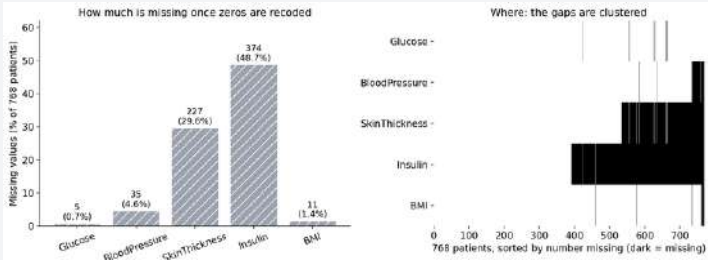
From the demo notebook (context lines omitted)

A • hidden zeros become missing values

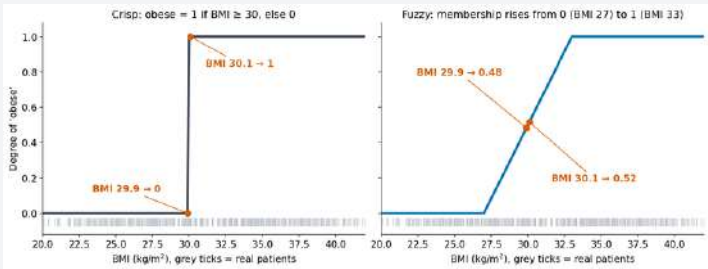
```
df[cols] = df[cols].replace(0, np.nan)
df[cols].isna().sum()
```

B • crisp threshold vs fuzzy membership

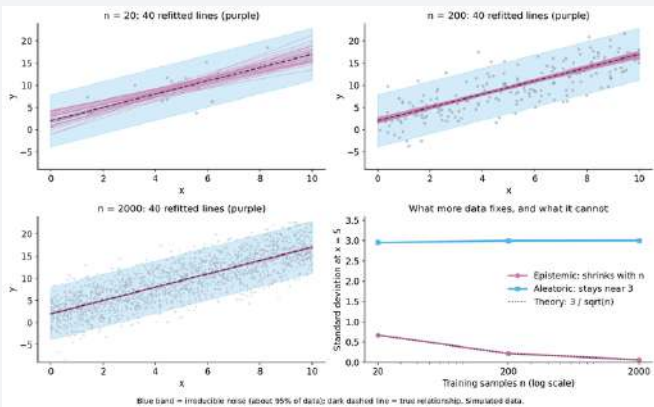
```
crisp = (bmi >= 30).astype(float)
fuzzy = np.clip((bmi - 27) / 6, 0, 1)
```



374 of 768 insulin values (48.7 %) were hidden zeros



BMI 29.9 gives crisp 0, fuzzy 0.48; BMI 30.1 gives crisp 1, fuzzy 0.52



Prediction spread 0.68, 0.22, 0.07 as n = 20, 200, 2,000; noise stays near 3

Live demo: real patient data, three questions

CS405 • PRESENTATION 1 • LIVE DEMO

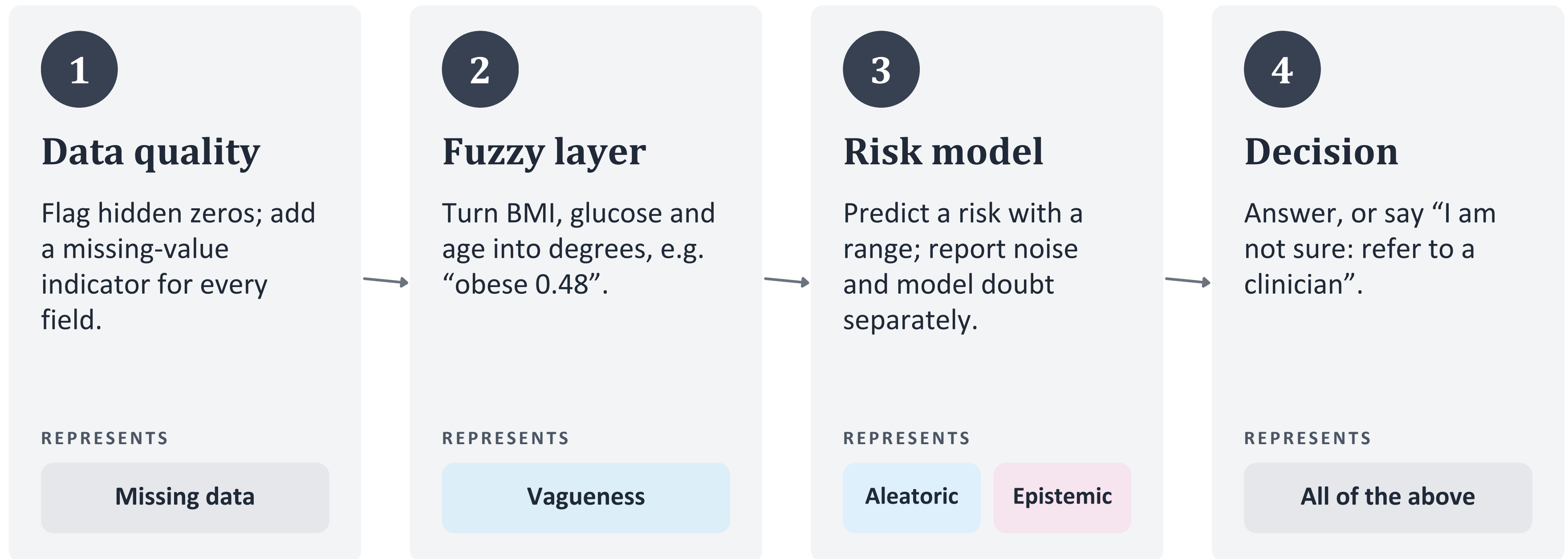


Scan to open the notebook in Google Colab

Advanced AI and Uncertainty • Week 1 workbook • Team 01

github.com/ahmedtayel714/Advanced-AI-and-Uncertainty--W1

Final-project idea: an uncertainty-aware diabetes-risk screener



Each kind of uncertainty gets its own component.

Five things to remember

- 1 Uncertainty comes in four kinds: randomness, vagueness, imprecision, missing data.
- 2 Aleatoric noise stays; epistemic doubt shrinks with data.
- 3 Crisp logic jumps; fuzzy sets give degrees; a degree is not a probability.
- 4 Missing data has mechanisms, and every fix has a price.
- 5 A trustworthy model can say “I am not sure” and knows why.

References

- [1] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Hoboken, NJ, USA: Pearson, 2021.
- [2] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.
- [3] A. P. Engelbrecht, *Computational Intelligence: An Introduction*, 2nd ed. Chichester, U.K.: Wiley, 2007.
- [4] E. H. Shortliffe, *Computer-Based Medical Consultations: MYCIN*. New York, NY, USA: Elsevier, 1976.
- [5] V. Gulshan et al., “Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs,” *JAMA*, vol. 316, no. 22, pp. 2402–2410, 2016, doi: 10.1001/jama.2016.17216.
- [6] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. 34th Int. Conf. Mach. Learn. (ICML)*, vol. 70, 2017, pp. 1321–1330.
- [7] National Transportation Safety Board, “Collision between vehicle controlled by developmental automated driving system and pedestrian, Tempe, Arizona, March 18, 2018,” Washington, DC, USA, Highway Accident Rep. NTSB/HAR-19/03, 2019. [Online]. Available: <https://www.nts.gov/investigations/AccidentReports/Reports/HAR1903.pdf>
- [8] National Transportation Safety Board, “Preliminary report: Highway HWY18MH010,” Washington, DC, USA, 2018.
- [9] E. Beede et al., “A human-centered evaluation of a deep learning system deployed in clinics for the detection of diabetic retinopathy,” in *Proc. CHI Conf. Human Factors Comput. Syst.*, 2020, doi: 10.1145/3313831.3376718.
- [10] E. Hüllermeier and W. Waegeman, “Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods,” *Mach. Learn.*, vol. 110, no. 3, pp. 457–506, Mar. 2021, doi: 10.1007/s10994-021-05946-3.
- [11] D. Hyde and D. Raffman, “Sorites paradox,” in *The Stanford Encyclopedia of Philosophy* (rev. Aug. 26, 2025). Accessed: Sep. 30, 2026. [Online]. Available: <https://plato.stanford.edu/entries/sorites-paradox/>
- [12] L. A. Zadeh, “Fuzzy sets,” *Inf. Control*, vol. 8, no. 3, pp. 338–353, Jun. 1965, doi: 10.1016/S0019-9958(65)90241-X.
- [13] D. B. Rubin, “Inference and missing data,” *Biometrika*, vol. 63, no. 3, pp. 581–592, 1976, doi: 10.1093/biomet/63.3.581.
- [14] J. W. Smith, J. E. Everhart, W. C. Dickson, W. C. Knowler, and R. S. Johannes, “Using the ADAP learning algorithm to forecast the onset of diabetes mellitus,” in *Proc. Symp. Comput. Appl. Med. Care*, 1988, pp. 261–265.
- [15] Plotly, “diabetes.csv (Pima Indians Diabetes Database),” plotly/datasets repository, GitHub. Accessed: Sep. 30, 2026. [Online]. Available: <https://github.com/plotly/datasets/blob/master/diabetes.csv>
- [16] World Health Organization, “Obesity and overweight,” fact sheet, Dec. 8, 2025. Accessed: Sep. 30, 2026. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight>

Thank You

Questions & Discussion

Matrouh University, Egypt

Presenter : Ahmed Ayman Tayel | Registration ID: 232611000001